

Boundary-Relative State-Time: When the Same Clock Time Does Not Mean the Same System History

Human-readable companion to Boundary-Relative State-Time: Measurement Lineage Across Nested System Boundaries

Human-readable companion to the completed scientific manuscript. This version preserves the scientific claims and limits of the manuscript while presenting the motivating problem and argument in more direct language. It does not constitute a separate research contribution.

Abstract

Two systems can exist for the same amount of clock time and still accumulate very different histories. One may cycle through large thermal, structural, electrical, chemical, or computational changes while another remains nearly unchanged. The scientific problem is not to replace ordinary time with a new universal clock. It is to ask how an accumulated history can be defined for a chosen system boundary, and what must remain invariant when that analytical boundary is enlarged. Boundary-relative state-time is a measurement-lineage framework for this problem. It requires the boundary, state variables, cross-boundary exchanges, accessible transitions, and accumulation rule to be declared before a history is interpreted. Its central rule is that when a larger boundary is treated as an extension of a smaller one at inherited resolution, the smaller system's previously assigned history and measurement meaning must remain recoverable. Otherwise the larger description is a different model lineage rather than the same measurement continued outward. The framework keeps direct whole-boundary accumulation distinct from component-preserving ledgers and from any optional scalar summary of those ledgers. It also separates boundary expansion from coarse-graining and from physically modifying the system. The mathematical ingredients are established; the residual contribution is the measurement-lineage discipline and its pre-outcome rejection rules. The framework earns scientific value only if a pre-specified implementation adds something beyond established domain metrics in prospective testing.

1. The intuitive problem: a clock tells you duration, not how much happened inside

Imagine two machines switched on at the same moment and shut down an hour later. One spends that hour nearly idle. The other heats, cools, switches states, moves loads, changes configuration, or repeatedly crosses stress thresholds. Ordinary clock time correctly says that both existed for one hour. It does not say that both accumulated the same internal history.

That distinction is the starting point of this paper. The historical intuition can be stated simply: external clock time does not tell you how long an internal system has aged alone. The final manuscript does not turn that intuition into a claim that physical time is subjective, that laboratory time is wrong, or that every system possesses one hidden intrinsic clock. Instead it asks a measurement question: if we want a coordinate for accumulated internal evolution, how do we keep that coordinate interpretable when we redraw the system boundary?

Existing science already provides many ways to measure accumulated change. Material-time approaches address aging in glasses and dissipative systems [1-4]. Thermodynamic length and information-geometric length measure path-dependent change in state spaces [5-7]. Dynamical activity and additive functionals provide other trajectory-level measures [8-10,32,33]. Boundary and compositional formalisms are also established [11,12,21,23-27,34-36]. Boundary-relative state-time does not claim ownership of those ingredients. Its narrower question is how to preserve the meaning of a previously defined history when the analytical boundary changes.

2. The boundary comes before the metric

Before asking how much a system changed, the analysis must say what counts as the system. A boundary can be physical, functional, or analytical. It need not describe a closed system. Energy, matter, information, and control may cross it in either direction. What matters is that the chosen boundary determines which changes are internal, which exchanges are external, and which variables belong to the state being measured.

This matters because the same physical event can change category when the boundary expands. Heat leaving an engine and entering a coolant loop is an external heat flow if the engine alone is the system. If the system is enlarged to include the coolant loop, that same transfer becomes internal redistribution. The physical event did not change. The bookkeeping did.

The same is true of control. A plant may look extraordinarily stable if the controller, sensing, computation, actuation, maintenance, and heat rejection that preserve that stability all sit outside the chosen boundary. Enlarging the boundary can reveal those costs. But the enlargement must not rewrite the history already assigned to the plant itself.

The manuscript therefore distinguishes analytical enlargement from two other operations. Coarse-graining changes resolution or compresses information. Physical modification changes the realized system by adding or removing components, controllers, materials, or couplings. Neither should be presented as though it were merely moving an analytical line around the same unchanged realization.

3. State-time is an accumulated-history coordinate, not a replacement for time

For a chosen boundary B, the analyst first declares a state record and an accumulation rule. The rule says which changes matter for the question, how they are scaled, and how they are accumulated. The general direct-scalar form is:

$$x_{i_B}^{\text{full}}[t_0, t_1] = A_B(\text{Gamma_B}[t_0, t_1]; M_B)$$

In plain language, the accumulated state-time of boundary B over an interval is whatever value the pre-specified measurement rule M_B assigns to the recorded history Gamma_B. The paper does not require one universal A_B. A thermal-aging application, an electrical-reliability application, and a computational-state application may legitimately use different rules.

For a direct scalar to count as a state-time coordinate in the manuscript's sense, it must be nonnegative, sensitive to the relevant history, and additive across adjacent intervals under a fixed boundary and fixed rule. 'History-sensitive' does not mean every distinct trajectory must receive a unique number. It means the measure need not be recoverable from the final state alone and may distinguish paths that share endpoints.

A geometric implementation is one possible realization: accumulate weighted path length through a declared state space. An event-count implementation is another: count specified transitions or threshold crossings. The framework deliberately permits both. The important scientific object is not the geometric equation itself but the declared lineage connecting boundary, recorded history, and measurement semantics.

The manuscript also blocks a common dimensional mistake. Energy can drive state change without being the same thing as state-time. Two systems can receive the same energy and follow different state trajectories because the delivery path, rate, environment, coupling, barriers, or accessible transitions differ. Joules remain joules. A state-history coordinate is not renamed energy.

4. The central rule: enlarging the boundary must preserve inherited history

Suppose B1 is contained inside a larger boundary B2. If B2 is genuinely an extension of the same analysis at inherited resolution, the B2 record must contain enough information to recover the B1 record. The manuscript writes this as a restriction relation:

$$\pi_{i_21}(\text{Gamma_B2}) = \text{Gamma_B1}$$

That equation is not a law of nature. It is the bookkeeping condition for calling the larger model an extension rather than a replacement. A transfer that is internal inside B2 may reappear as an external exchange when the record is restricted back to B1, but the underlying physical event and inherited state history must agree.

For three nested boundaries B1 subset B2 subset B3, direct and staged restriction must also agree:

$$\pi_{i_ii} = \text{id}, \quad \pi_{i_31} = \pi_{i_21} \circ \pi_{i_32}$$

A whole-system description therefore cannot yield one history for B1 when restricted directly and a different history when restricted through B2. This coherence pattern is standard mathematics rather than new mathematics [11,12]. Its role here is to make the measurement-lineage requirement operational.

Measurement rules have to respect the same lineage. If enlarging the boundary silently changes the scale, weight, event definition, normalization, or observational convention previously used for the inherited subsystem, then the analyst has changed models. That may be scientifically legitimate, but it must be labeled as a new measurement lineage rather than attributed to boundary movement itself.

5. Three objects that should not be collapsed into one number

The manuscript separates three kinds of accounting because they answer different questions.

Object	Plain-language role	What it is not
Direct full-boundary accumulation (ξ_B^{full})	Apply one declared rule directly to the complete state description of the chosen boundary.	Not automatically equal to the sum of subsystem histories.
Component-preserving ledger (Ξ_B)	Keep the declared histories of contained components and relevant exchanges recoverable when the boundary expands.	Not required to be a scalar.
Optional scalarization (σ_B)	Summarize a heterogeneous ledger only when a justified aggregation rule is declared.	Not a universal total history and not interchangeable with direct full-boundary accumulation.

This separation matters whenever coupling creates cross terms or heterogeneous histories that cannot be faithfully reduced by simple addition. If the larger system's direct trajectory contains interaction terms, its directly measured path length can differ from the arithmetic sum of separately accumulated component path lengths. That is not an inconsistency. It means the two quantities answer different questions.

The ledger is therefore the safest primary composite object. It can preserve different native histories - path lengths, event counts, damage records, maximum excursions, or other domain-specific objects - without pretending they share one natural unit. A scalar summary is allowed only when the aggregation semantics are explicit and coherent across boundaries.

6. Why endpoints are not enough

Two trajectories can begin and end at the same states yet accumulate different histories. A system that moves away from an initial state and returns can have zero net displacement while having undergone substantial path-dependent change. This is why the framework does not identify accumulated history with endpoint difference.

The same logic applies to threshold events. Two records can have similar average levels and the same final state while one repeatedly crosses a stress threshold and the other never does. If the domain says threshold crossings matter, an event-count rule may be more appropriate than geometric path length. The framework is intentionally metric-plural: the rule should be chosen and justified from the domain, then tested prospectively rather than fitted after the outcome is known.

Sampling resolution also has to be declared. Arbitrarily fine sampling of a noisy or Brownian-like trajectory can make ordinary geometric path length diverge. A valid implementation must therefore specify the observational scale, filtering, event definition, or stochastic construction that gives the measurement stable meaning. A reported state-time that changes only because the same physical record was oversampled is not admissible under the proposed discipline.

7. What boundary expansion can reveal without rewriting the subsystem

The engine-and-cooling-loop example makes the boundary rule intuitive. At a narrow boundary, coolant can look like an external sink. At a wider boundary, the same heat transfer is internal, and radiator or environmental exchange becomes the external flow. The larger description can reveal additional history - pump work, coolant heating, fan control, heat rejection - while preserving the engine history already measured at the smaller boundary.

The manuscript's LED thermal-coupling example uses public manufacturer data to illustrate a similar reclassification problem. Thermal coupling between emitters means a narrow component view can hide contributions that become visible in a wider fixture-level account. The example is explicitly illustrative. It does not validate a state-time metric or claim new device physics.

This distinction is important for systems claims that appear to create stability, efficiency, or low variation locally. If the apparent improvement is maintained by work performed elsewhere, widening the boundary can expose that burden. But the new burden is added history, not a license to revise what the narrower subsystem previously did.

8. What the framework can reject before outcome fitting

A measurement framework has scientific value only if it can rule out some constructions before seeing the desired result. Boundary-relative state-time imposes several such constraints.

- A larger-boundary model fails the extension test if the inherited subsystem history cannot be recovered at inherited resolution.
- A nested hierarchy is incoherent if direct restriction and staged restriction yield different inherited histories.
- A claimed boundary effect is misclassified if the analyst changed the measurement rule, reference scale, event definition, or resolution without declaring a new model lineage.
- A scalar comparison across boundaries is not justified merely because each boundary produces a number; aggregation semantics must themselves restrict coherently.
- A direct scalar that depends on arbitrary oversampling rather than the declared physical history is not admissible.
- If a standard domain metric predicts the target outcome as well as or better than the proposed implementation, the standard metric should win.

These are measurement-lineage failures rather than metaphysical failures. They do not say the physical system is impossible. They say a particular history claim has not preserved the meaning required to compare nested boundaries as one continuing analysis.

9. What established science already owns

The paper's formal ingredients are intentionally conservative. Intrinsic/material time, thermodynamic and information geometry, additive functionals, boundary energy accounting, compositional system formalisms, measurement invariance, provenance, coarse-graining, and hierarchical refinement all have substantial prior art [1-13,21-36]. The manuscript does not claim that changing the boundary is a new discovery or that restriction maps are a new mathematical invention.

The residual contribution is the assembly of those ideas into one measurement rule for accumulated histories: declare the boundary and semantics first; preserve component histories under analytical enlargement; distinguish direct whole-boundary accumulation from ledger aggregation; require restriction coherence; and treat a changed inherited rule as a new model lineage. That claim is methodological. It does not establish a universal physical clock, a universal scalar of aging, or a new fundamental time variable.

10. What would make the proposal shrink or disappear

The manuscript reserves empirical usefulness for prospective tests. An implementation should be pre-specified and compared against the strongest established domain measures. If boundary-relative state-time adds no predictive, explanatory, audit, or consistency value beyond ordinary domain accounting, there is no reason to retain it as an additional scientific object.

Likewise, if every useful application reduces cleanly to existing provenance, measurement-invariance, open-system, or compositional modeling practice without the measurement-lineage architecture changing any inference, the residual contribution is subsumed. That would not make the component ideas false. It would mean the higher-level framework is redundant.

The manuscript is also cautious about biological and cognitive applications. Such systems may eventually be relevant because their histories depend strongly on environment and nested control, but they introduce severe

observability and identifiability problems. The paper explicitly recommends validating the framework first in cleaner systems with better measurement and repeatability.

11. Conclusion

The central idea can be stated without the formalism: clock time tells us how long an interval lasted, but not how much of a chosen system's relevant internal history accumulated during that interval. If we construct an accumulated-history coordinate, its meaning depends on the analytical boundary and the measurement rule.

When the boundary expands, new histories can appear and old exchanges can be reclassified. What cannot happen silently is for the inherited subsystem's already-defined history to acquire a different meaning while the analyst still calls the larger model an extension of the same measurement. Direct and staged recovery must agree, and any optional scalar aggregation must preserve its own semantics.

Boundary-relative state-time therefore contributes a discipline of measurement lineage, not a second physical time. Its success depends on whether pre-specified implementations survive comparison with established metrics and make nested-boundary histories more coherent, traceable, or predictive. If they do not, the framework should give way to the established methods that already do the job.

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